Exploring variation in the exponential metalaw of physics

Work in progress: this writeup is still being updated.

Abstract

Previous work has found that the ranked frequencies of mathematical operators in physics equations approximately follow an exponential law. The fitted decay rate for Encyclopaedia Inflationaris is lower than those of several broader physics collections. Proposed explanations concern the conventions used to write the equations, the inflationary collection's focus on one physical concept, its hypothetical models, and the energy scales involved. We revisit the fitted difference using log least squares and estimate uncertainty by resampling whole equations. We examine the proposed explanations through changes to constant conventions and comparisons with modified-gravity models and physics branches. Setting $c = \hbar = 1$ in the textbook expressions and removing the common amplitude prefactor from the inflationary potentials leave most of the fitted difference intact. A comparison collection from modified gravity, which also contains hypothetical models focused on one physical concept, gives a decay rate close to the inflationary value. The fitted rates vary across seven branches of the expanded Feynman collection, with some estimates changing substantially when ambiguous chapters are reassigned. These comparisons document variation in operator-frequency patterns across the selected collections, while leaving its physical explanation unresolved.

1 Introduction

Previous work [1] has shown that the ranked frequencies of mathematical operators in physics equations can be described by an exponential law, in contrast to Zipf's power law for word frequencies in natural language. Each equation is represented as an expression tree, and occurrences of each operator type are counted across a collection of equations. The operator types are then ranked from most to least frequent. The counting convention includes variables, constants and numerical literals alongside arithmetic operations and named functions.

The exponential model takes the form

$$p_r(\beta) = \frac{e^{-\beta r}}{\sum_{s=1}^{R} e^{-\beta s}}, \qquad r = 1, \ldots, R, \tag{1}$$

where R is the number of operator types present. The decay rate β describes how rapidly frequency decreases with rank. A larger value corresponds to a more concentrated distribution, in which the most frequent types account for a greater share of occurrences.

The original study found similar fitted decay rates for the Feynman benchmark, Wikipedia equations and the Cambridge Handbook of Physics Formulas. Encyclopaedia Inflationaris gave a lower rate [1]. The proposed explanations concerned the treatment of physical constants, the collection's focus on one physical concept, the hypothetical nature of its models, and the energy scales involved. These possibilities motivate comparisons of both the way the equations are written and the kinds of physics represented in each collection.

This project revisits the fitted difference and examines these proposed explanations. We estimate decay rates by log least squares and use whole-equation resampling to obtain individual bootstrap intervals. Applying the same procedure across the collections allows us to compare their fitted rates and assess how much those estimates vary when the selected equations change.

We first examine constant conventions. Setting $c = \hbar = 1$ removes explicit factors of the speed of light and the reduced Planck constant from textbook expressions. Removing the common amplitude prefactor M^4 from the inflationary potentials makes a corresponding change to that collection. Such rewritings can change the operator counts even when they describe the same physics. We measure how much these particular changes alter the fitted difference.

Encyclopaedia Inflationaris assembles alternative models of single-field slow-roll inflation, expressed through proposed forms of the inflaton potential [2]. The collection therefore focuses on one physical quantity and

consists of hypothetical models proposed to describe inflation. The broader textbook collections cover many quantities and physical topics. This difference in scope may affect which operators occur and how frequently they are used.

To explore this possibility, we assemble a comparison collection from $f(R)$ gravity. In these theories, the gravitational action contains a function of the scalar curvature R, and different models propose different forms of this function [3]. Like the inflationary collection, this collection contains hypothetical models focused on one physical concept. It allows us to ask whether another collection with these features has a similar fitted rate, although the comparison cannot separate the effects of the two features. Section 2 and Appendix A identify the papers used to assemble the collection and the source of each expression.

The proposed dependence on energy scale also motivates a comparison across branches of physics. We expand the 100-expression AI Feynman benchmark [4] with expressions from all three volumes of the Feynman Lectures [5]. Comparing branches within a single source reduces differences arising from notation and editorial conventions. We also examine alternative assignments for chapters that span more than one branch. These comparisons explore variation with physical domain, but do not isolate energy scale from other differences between the equations.

This project makes three contributions. First, it provides a reproducible reassessment of the fitted decay rates with individual bootstrap uncertainty estimates. Second, it assembles an expanded Feynman collection with source references and branch labels, together with a catalogue of 37 modified-gravity expressions. Third, it examines how the fitted rates vary with constant conventions, the scope of the collections and the assignment of equations to physics branches.

Section 2 describes the collections, operator counting, fitting procedure and uncertainty estimates. Section 3 revisits the inflationary difference through changes to constant conventions and the modified-gravity comparison. Section 4 examines variation across physics branches and sensitivity to their classification. Section 5 discusses the interpretation and limitations of these comparisons, including their possible relevance to symbolic regression. Section 6 summarises what the comparisons establish and which explanations remain unresolved. The appendices document catalogue sources, sensitivity checks, bootstrap calibration and the exponential limit of the Zipf–Mandelbrot distribution.

2 Data and methods

2.1 Equation collections

We begin with three collections released with the original study [1]: 100 expressions from the AI Feynman benchmark, 41 equations from Wikipedia's list of scientific equations named after people [6], and 70 inflaton potentials from Encyclopaedia Inflationaris [2]. We extend the Wikipedia collection with 35 algebraic expressions, bringing its total to 76. As a mathematical comparison, we also include 61 probability density or mass functions from Leemis and McQueston [7].

For the comparison across physics branches, we expand the Feynman benchmark using expressions from all three volumes of the Feynman Lectures [5]. After selection and deduplication, this adds 916 expressions. The combined collection contains 1016 expressions, of which 968 are assigned to the seven branches considered here. Because the original benchmark is part of the expanded collection, the two collections overlap.

The modified-gravity collection contains 37 proposed curvature functions from the review and model papers listed in Appendix A. These expressions include related variants: 17 come from the model families examined by Pérez-Romero and Nesseris [8]. The collection therefore contains 37 expressions, which should not be interpreted as 37 independent model families. Each catalogue entry records its expression and source, and Appendix A provides the corresponding source table.

2.2 Equation selection and operator counting

For the additional Feynman Lectures expressions, we select equations that give one quantity as an explicit algebraic expression. We count operators on the right-hand side. We exclude expressions consisting of a single symbol or number, as well as those containing derivatives, integrals, sums, limits, operator hats or complex conjugates.

Each expression is represented as a tree, following the counting conventions of the original study [1]. Arithmetic operations and named functions contribute operator occurrences. Variables, physical constants and nu-

merical literals are also counted, each as a separate category of tree leaves. Powers x^2, x^3 and x^4 are represented by distinct unary operators. Appendix A gives further transcription and counting conventions.

We assign expressions from the Lectures to physics branches according to their source chapters. Benchmark expressions retain their existing labels, apart from the relativistic-electrodynamics adjustment described in Appendix A. Combining fluid and continuum mechanics gives seven branches: classical mechanics, special relativity, electromagnetism, waves and optics, thermodynamics and kinetic theory, quantum mechanics, and fluid and continuum mechanics.

The other 48 expressions remain in the expanded collection but are excluded from the seven-branch comparison. They comprise 31 mathematics expressions, nine probability expressions, six general-relativity expressions and two without a branch assignment. Some chapters span more than one branch; we examine alternative assignments for these chapters as a sensitivity check.

2.3 Fitting the exponential decay rate

For each collection or branch, we count occurrences of every operator type and arrange the positive counts in descending order,

$$n_1 \geq n_2 \geq \cdots \geq n_R > 0.$$

The observed frequency at rank r is

$$f_r = \frac{n_r}{N}, \qquad N = \sum_{r=1}^{R} n_r.$$

We fit the normalised exponential model introduced in Eq. (1) by minimising the squared differences between observed and model log frequencies:

$$Q(\beta) = \sum_{r=1}^{R} \left[\log f_r - \log p_r(\beta)\right]^2 . \tag{2}$$

Each observed rank receives equal weight in this criterion. We include every operator type with a positive count. Types absent from a collection are omitted because their log frequencies are undefined.

The ranking is constructed separately for each collection or branch. Consequently, both the number of observed types and the operator occupying a particular rank can differ between groups. The fitted rate $\hat{\beta}$ summarises how concentrated each ranked distribution is; it does not describe which operators are most common.

We constrain the numerical optimisation to $0 \leq \beta \leq 5$ and flag fits near either boundary. Calculating this fitted summary does not require operator occurrences within an equation to be independent. The assumptions needed to interpret its uncertainty enter through the resampling procedure below.

2.4 Uncertainty from equation resampling

We estimate uncertainty by resampling whole equations within each collection or branch. Each bootstrap sample contains the same number of equations as the original group, drawn with replacement. An equation can therefore appear more than once in a sample, while another may be absent.

For each sample, we recount the operators, determine which types are present, rank their frequencies and refit β. Repeating this procedure 10000 times gives a distribution of fitted rates. We report the fit to the original collection, the standard deviation of the bootstrap estimates as a standard error, and their 2.5th and 97.5th percentiles as an individual 95% bootstrap interval.

Resampling whole equations keeps together the operators that occur within each expression. Interpreting the resulting intervals as sampling uncertainty assumes that the equations are sufficiently independent and represent a wider collection of comparable expressions. Related equations or variants of the same model can limit this assumption. The intervals also do not account for systematic omissions or transcription errors in the collections.

The 95% level is approximate. Its accuracy depends on how well resampling the observed equations represents variation in the fitted rate. We report intervals for each rate individually. The synthetic checks in Appendix C examine their coverage and show that it can fall below the nominal level.

We also resample whole source chapters to examine dependence among equations from the same chapter. This check, together with alternative branch assignments and selection rules, is described in Appendix B.

For reproducibility, we use random seed 20261013 and calculate percentiles by linear interpolation (Hyndman–Fan type 7). Failed and degenerate fits are recorded, and intervals affected by missing fits are marked unresolved.

3 Revisiting the inflationary difference

3.1 Comparison of the fitted rates

We first examine whether the lower decay rate reported for Encyclopaedia Inflationaris is reproduced using the fitting procedure described in Section 2. Table 1 and Figure 1 summarise the estimates for all collections.

The Feynman benchmark gives a fitted rate of $\hat{\beta} = 0.342$, with an individual 95% bootstrap interval of $[0.319, 0.371]$. Encyclopaedia Inflationaris gives $\hat{\beta} = 0.269$, with an interval of $[0.253, 0.293]$. The point estimates differ by 0.073, and their individual intervals do not overlap. The lower inflationary estimate is therefore reproduced with log least squares and equation resampling.

A lower decay rate corresponds to a slower decline in frequency with rank. Under the fitted exponential model, operator occurrences are therefore less concentrated among the most frequent types in the inflationary collection than in the Feynman benchmark. This describes a difference between their ranked distributions, without identifying its physical origin.

Corpus	Expr.	Instances	Ranks	$\hat{\beta}$	Boot. SE	Individual interval
Feynman benchmark	100	1167	19	0.342	0.013	[0.319, 0.371]
Feynman benchmark + Lectures	1016	10693	22	0.354	0.008	[0.333, 0.366]
Wikipedia	76	1006	17	0.315	0.015	[0.284, 0.346]
Encyclopaedia Inflationaris	70	1376	25	0.269	0.010	[0.253, 0.293]
Leemis	61	1044	24	0.269	0.014	[0.250, 0.304]
f(R)	37	650	24	0.270	0.021	[0.264, 0.347]

Table 1: Corpus estimates from log least squares. Intervals are nominal individual 95% equation-bootstrap intervals. The instances column counts all operator occurrences, including nullary types.

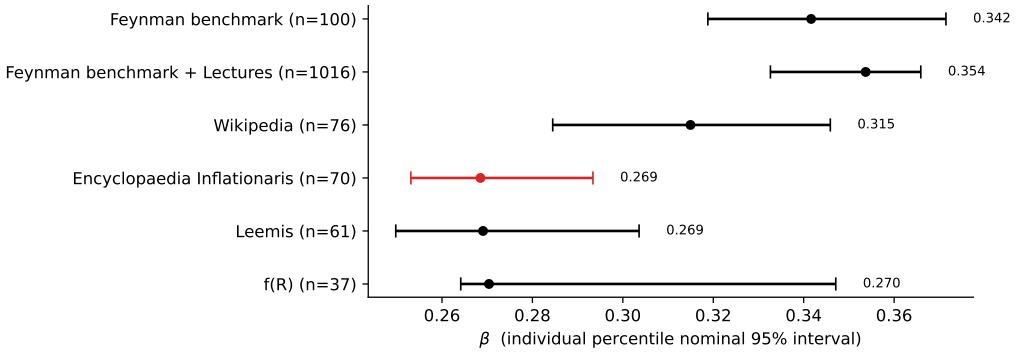


Figure 1: Corpus estimates and nominal individual 95% equation-bootstrap intervals.

3.2 Changes to constant conventions

One proposed explanation concerns the conventions used to write the equations. Physical constants and amplitude prefactors contribute to the expression trees, so removing them can change the operator counts and the fitted decay rate. We examine two such changes.

First, we set $c = \hbar = 1$ in the textbook expressions. This affects 32 expressions in the Feynman benchmark and lowers its fitted rate from 0.342 to 0.332. Only two expressions in the Wikipedia collection are affected; its rate rounds to 0.315 both before and after the change.

Second, we remove the common amplitude prefactor M^4 from the inflationary potentials by setting $M = 1$. This affects all 70 potentials and lowers the fitted rate from 0.269 to 0.261.

After both changes, the difference between the Feynman benchmark and the inflationary collection is 0.071, compared with 0.073 before the changes. These conventions therefore account for little of the observed difference in point estimates. Other choices of algebraic representation may also affect operator counts, but they are not examined in this comparison.

3.3 Comparison with modified gravity

We next consider the proposed explanations involving the inflationary collection's focus on one physical concept and its inclusion of hypothetical models. The $f(R)$ gravity collection shares these features: its expressions are proposed forms of a curvature function entering the gravitational action.

The 37 modified-gravity expressions give a fitted rate of $\hat{\beta} = 0.270$, with an individual 95% bootstrap interval of $[0.264, 0.347]$. Its point estimate is close to the inflationary value of 0.269, and the intervals overlap. A similarly low fitted rate is therefore found in another collection of hypothetical models centred on one physical concept.

This similarity does not establish why the rates are close. Both features change together in the comparison, so their effects cannot be distinguished. The modified-gravity collection also contains related variants of model families. These variants influence the operator frequencies and limit the independence assumption underlying equation resampling.

Together, the comparisons reproduce the lower inflationary estimate and show that it changes little relative to the Feynman benchmark under the tested constant conventions. The modified-gravity collection provides a further example of a similar fitted rate, while leaving its explanation unresolved. We next examine variation across physics branches within the expanded Feynman collection.

4 Variation across physics branches

The original study proposed differences in energy scale as one possible explanation for the lower inflationary decay rate [1]. Inflation also concerns a different branch of physics from much of the material in the broader equation collections. This motivates us to examine whether fitted decay rates vary across physics branches. We use the expanded Feynman collection because the expressions in the Feynman Lectures are publicly accessible [5]. Branch labels provide a way to explore variation with physical domain, although they do not isolate the effect of energy scale.

The fitted rates show a pattern suggestive of two broad groups (Table 2 and Figure 2). Quantum mechanics, electromagnetism, waves and optics, and classical mechanics have estimates between 0.325 and 0.360, with overlapping individual bootstrap intervals. Thermodynamics and kinetic theory and fluid and continuum mechanics have higher estimates, at 0.403 and 0.422, and their intervals also overlap.

The separation between these groups is incomplete. Special relativity has a relatively wide interval that overlaps estimates in both groups. The plot therefore suggests a grouping of the fitted rates, without establishing a sharp division between branches. Figure 2 also includes the original Feynman benchmark, the full expanded Feynman collection, and the modified-gravity, inflationary and Wikipedia collections for comparison.

Branch	Expr.	Instances	Ranks	$\hat{\beta}$	Boot. SE	Individual interval
Special relativity	89	1073	14	0.382	0.028	[0.322, 0.432]
Quantum mechanics	221	2374	17	0.327	0.016	[0.286, 0.349]
Electromagnetism	312	3285	18	0.325	0.010	[0.313, 0.353]
Waves and optics	126	1448	15	0.346	0.017	[0.303, 0.367]
Classical mechanics	105	1014	15	0.360	0.020	[0.297, 0.377]
Thermodynamics and kinetic theory	73	727	16	0.403	0.021	[0.371, 0.452]
Fluid and continuum mechanics	42	360	13	0.422	0.027	[0.377, 0.483]

Table 2: Branch estimates and nominal individual 95% equation-bootstrap intervals.

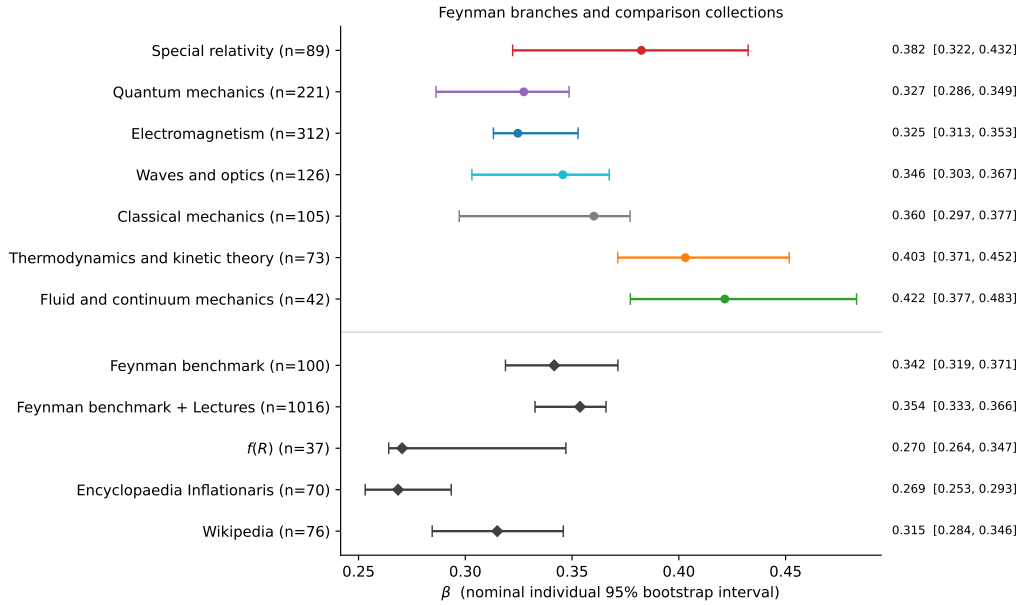


Figure 2: Fitted rates for the seven Feynman branches (circles) and five comparison collections (diamonds, below the dividing line), with nominal individual 95% equation-bootstrap intervals. The comparison collections are the 100-expression Feynman benchmark, the full expanded Feynman collection, $f(R)$ gravity, Encyclopaedia Inflationaris and Wikipedia. The benchmark is contained in the expanded collection, whose expressions also supply the branch estimates.

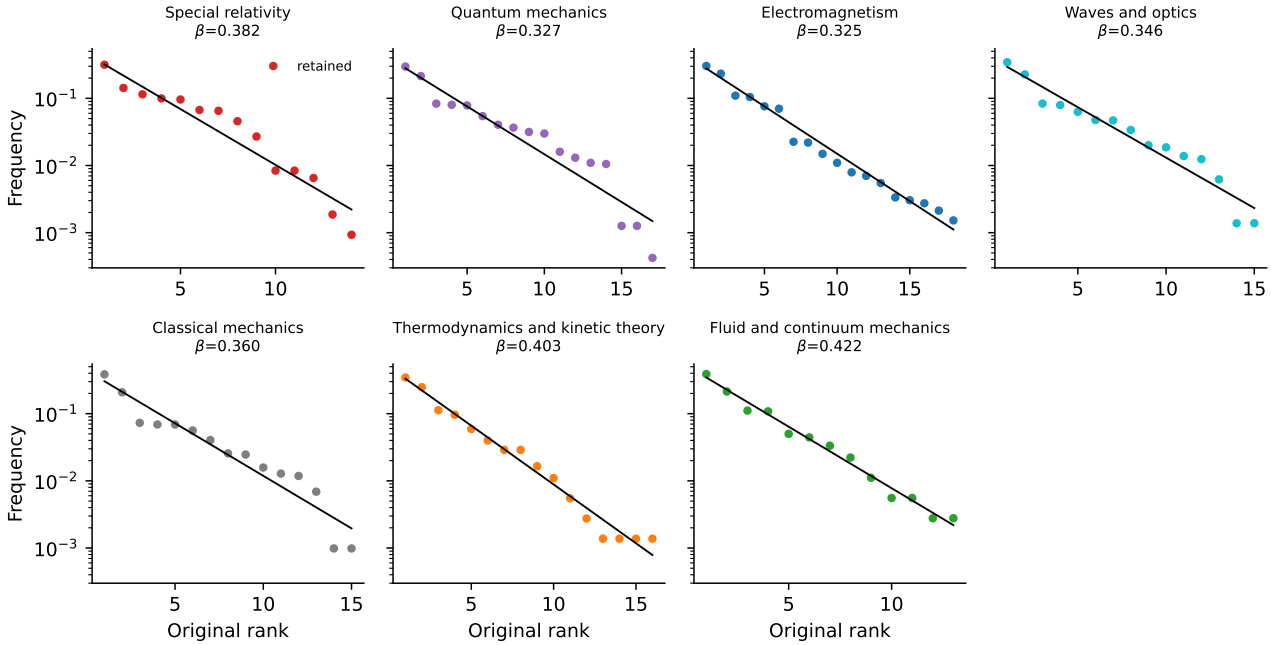


Figure 3: Observed rank–frequency distributions and fitted exponentials for the seven branches.

Figure 3 shows the distributions underlying these estimates. Each branch is ranked separately, and both the observed operator types and their ordering can differ. The fitted rates describe how concentrated the ranked frequencies are; identifying the operators responsible for these patterns requires examining their identities separately.

Additional checks examine the effects of branch classification and relationships between equations from the same chapter. The special-relativity estimate changes substantially when ambiguously classified chapters are removed or reassigned. Resampling whole chapters widens the quantum- and fluid-mechanics intervals, while leaving them separated. Appendix B gives the procedures and results. These checks help distinguish patterns in the selected expressions from conclusions that would require broader evidence about each physics branch.

5 Discussion

5.1 What the fitted rate describes

The fitted β is a one-parameter summary of a ranked operator distribution under the stated criterion. Ranking discards operator identities: two collections can have similar fitted rates while using different operators. Conversely, variation in β describes a difference in the concentration of the ranked frequencies, without identifying which physical or representational choices produced it.

The observed operator sets also differ between branches. Because each fit uses the types present in its own group, the comparisons reflect both the observed vocabulary and its relative frequencies.

5.2 Corpus construction and uncertainty

The collections consist of selected expressions. Inclusion criteria, algebraic representation, branch labels and the number of variants retained from a model family can all affect the resulting counts. The branch-assignment checks demonstrate one such effect. A single source reduces some differences in notation and editorial choices, but cannot make the selected expressions representative of every equation in a branch.

Equation resampling measures variation when drawing comparable equations from the observed collection. It preserves dependence within equations, but does not capture systematic selection or transcription errors. Chapter resampling examines one source of dependence between equations. In the synthetic calibration checks, the individual intervals contained the population rate in 74.5% to 94.5% of datasets across the tested branches and scenarios, below the nominal 95% level. We therefore treat the reported intervals as approximate uncertainty estimates (Appendix C).

5.3 Physical interpretation and symbolic regression

The lower inflationary estimate persists after the specified constant changes, while the expanded modified-gravity collection has a similar estimate to inflation. These observations leave the physical explanation unresolved. In particular, the branch comparison does not isolate energy scale from other differences between the equations.

Operator statistics from individual domains could nevertheless be useful when constructing priors for symbolic regression. The relevant prior might describe operator identities and combinations as well as the rank–frequency decay rate. Whether such domain-specific priors improve equation recovery requires a separate evaluation.

6 Conclusion

We revisit the exponential metalaw using log least squares and individual equation-bootstrap intervals. The lower fitted rate of Encyclopaedia Inflationaris relative to the Feynman benchmark is reproduced, and the specified changes to constant conventions leave most of the difference intact. The expanded modified-gravity collection gives a rate close to the inflationary value. Within the Feynman collection, the fitted rates vary across branches, although branch assignment affects the estimates and the intervals are approximate. The project documents these patterns and their sensitivity to the construction of the collections, while leaving their physical explanation open.

A Catalogue sources and transcription conventions

The catalogue records the expression and its source reference. Lectures entries include volume, chapter and equation identifiers. The additional Wikipedia entries follow the original release cutoff in the same alphabetical

list. The source references and counting metadata are retained with the individual records.

Physical constants such as c, $\hbar$ and G are classified as constants. The symbol q_e retains its classification as a variable in the benchmark. Time averages written as $\langle x \rangle$ are represented by a single symbol. In the modified-gravity catalogue, R denotes curvature rather than the gas constant, and model parameters are counted as variables. The catalogue stores the full curvature function: when a source writes $F(R) = R + f(R)$, the correction $f(R)$ is combined with R. Optional additional terms proposed later in a source are not included in the defining expression unless that expression itself contains them.

For branch assignment, the benchmark expression from the relativistic-electrodynamics chapter in Volume I is assigned to special relativity. The corresponding Lectures chapters—Volume I, Chapter 34, and Volume II, Chapters 25, 26 and 28—receive the same classification. The catalogue records ambiguous assignments for the sensitivity analysis.

Table 3 identifies the source used to transcribe each modified-gravity expression. Where a review is cited, it is the transcription source; the citation does not attribute the original proposal of every model to that review. The entry identifiers match those in the catalogue.

Table 3: Transcription sources for the 37 modified-gravity expressions.

Catalogue entry	Source
`fr.power`	[3]
`fr.inverse_power`	[3]
`fr.quadratic`	[3]
`fr.inverse_plus_quadratic`	[3]
`fr.chameleon`	[3]
`fr.logarithm`	[9]
`fr.exponential`	[10]
`fr.sine`	[3]
`fr.two_scales`	[3]
`fr.starobinsky_2007`	[3]
`fr.hu_sawicki`	[11]
`fr.tsujikawa`	[12]
`fr.exponential_gravity`	[13]
`fr.appleby_battye`	[14]
`fr.miranda_log`	[15]
`fr.rational_nojiri`	[16]
`fr.two_tanh`	[16]
`fr.odd_rational`	[16]
`fr.kruglov_arctan`	[17]
`fr.kruglov_arctan_quadratic`	[18]
`fr.pn_sqrt`	[8]
`fr.pn_inverse`	[8]
`fr.pn_three_halves`	[8]
`fr.pn_arctanh`	[8]
`fr.pn_sin`	[8]
`fr.pn_sinh`	[8]
`fr.pn_exp`	[8]
`fr.pn_tanh`	[8]
`fr.pn_asin`	[8]
`fr.pn_log`	[8]
`fr.pn_tan`	[8]
`fr.pn_arctan`	[8]
`fr.pn_erf`	[8]
`fr.pn_asinh`	[8]
`fr.pn_polynomial`	[8]
`fr.pn_pade`	[8]

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Table 3 (continued)

Catalogue entry	Source
`fr.pn_hypergeometric`	[8]

B Sensitivity checks

B.1 Source-chapter resampling

We resample whole chapters jointly across branches. Each chapter is assigned to a stratum corresponding to the branch containing most of its retained equations, with ties resolved using the recorded branch order. Within each stratum, we sample the original number of chapters with replacement. Every equation from a selected chapter receives the same multiplicity, including equations assigned to other branches. This preserves dependence induced by the 11 chapters represented in more than one branch; the number of equations per branch can vary between replicates.

The individual chapter-bootstrap interval is [0.375, 0.523] for fluid and continuum mechanics and [0.279, 0.356] for quantum mechanics. Both are wider than their equation-bootstrap intervals. The four chapters represented in fluid and continuum mechanics provide a small number of resampling units, so this check gives limited information about its uncertainty.

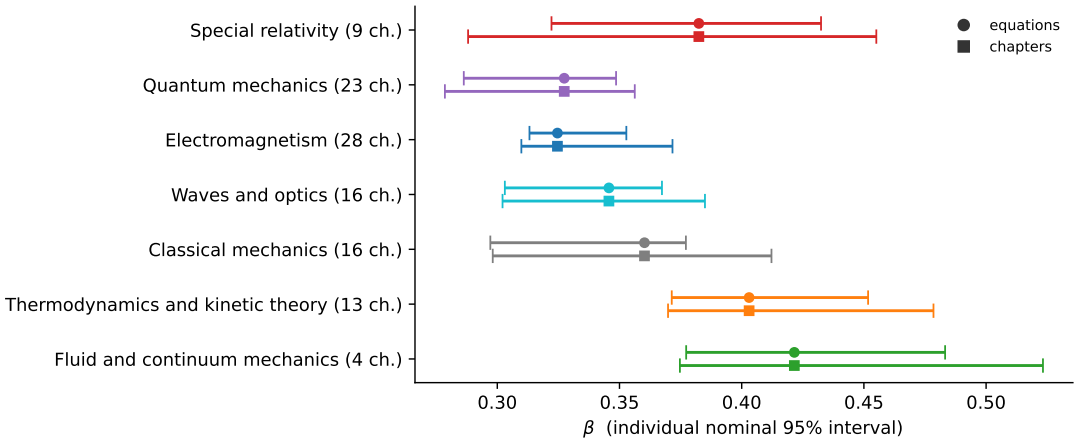


Figure 4: Equation and joint chapter bootstrap intervals at the nominal individual 95% level. Labels give the number of source chapters represented in each branch.

B.2 Branch assignments and selection

The alternative specifications remove chapters with ambiguous assignments, reassign those chapters, remove expressions flagged for averages or imaginary units, or remove Volume III. They use the same log-least-squares fitting procedure. Table 4 reports the estimates and individual equation-bootstrap intervals; the original branch assignment remains the main analysis.

Specification	Branch	Expr.	$\hat{\beta}$	Individual interval
Ambiguous chapters removed	SR	32	0.231	[0.204, 0.289]
	QM	151	0.320	[0.284, 0.350]
	EM	198	0.315	[0.302, 0.351]
	Waves	48	0.291	[0.265, 0.341]
	Classical	75	0.354	[0.298, 0.390]
	Thermo	61	0.405	[0.368, 0.461]
	Fluid	42	0.422	[0.377, 0.483]
Ambiguous chapters reassigned	SR	32	0.231	[0.204, 0.289]
	QM	182	0.335	[0.299, 0.364]
	EM	343	0.336	[0.315, 0.359]
	Waves	63	0.303	[0.283, 0.345]
	Classical	75	0.354	[0.298, 0.388]
	Thermo	89	0.353	[0.321, 0.388]
	Fluid	44	0.419	[0.374, 0.478]
Notation-flagged expressions removed	SR	88	0.395	[0.314, 0.437]
	QM	164	0.317	[0.284, 0.341]
	EM	273	0.330	[0.318, 0.361]
	Waves	99	0.362	[0.320, 0.393]
	Classical	97	0.365	[0.309, 0.386]
	Thermo	69	0.400	[0.368, 0.449]
	Fluid	39	0.422	[0.380, 0.500]
Volume III removed	SR	89	0.382	[0.321, 0.431]
	QM	26	0.349	[0.319, 0.426]
	EM	312	0.325	[0.313, 0.353]
	Waves	126	0.346	[0.303, 0.368]
	Classical	105	0.360	[0.296, 0.377]
	Thermo	73	0.403	[0.370, 0.451]
	Fluid	42	0.422	[0.378, 0.481]

Table 4: Alternative branch assignments and selection rules. Each entry reports the fitted rate and its nominal individual 95% equation-bootstrap interval. Waves denotes waves and optics; fluid denotes fluid and continuum mechanics.

C Bootstrap calibration

We examine interval coverage using 200 synthetic datasets in each of three scenarios, each with seven branches and 400 bootstrap replicates. All scenarios have population marginal operator frequencies following an exponential with rate $\beta = 0.35$. The first uses identical distributions and independent equation count vectors. The second varies equation counts, operator vocabularies and expression lengths, with dependence within an equation introduced through a shared random probability vector. The third introduces a shared random probability vector within each chapter and resamples whole chapters, including a branch with only four chapters.

For each branch, we record how often the individual percentile interval contains the population rate. Coverage across the tested branches and scenarios ranges from 74.5% to 94.5%, with the lowest value in the chapter-dependent scenario. The nominal 95% intervals therefore do not consistently achieve that coverage in these checks. We also record failed fits and binomial Monte Carlo intervals for the coverage estimates. The synthetic count vectors do not enforce the grammar of expression trees. These simulations diagnose limitations in the specified cases and do not provide a calibration correction for the observed corpora.

D Exponential limit of the Zipf–Mandelbrot distribution

For a fixed set of ranks $r = 1, \dots, R$, the normalised Zipf–Mandelbrot distribution is

$$p_r(\alpha, b) = \frac{(r+b)^{-\alpha}}{\sum_{s=1}^{R}(s+b)^{-\alpha}}, \qquad \alpha > 0, \quad b > 0. \tag{3}$$

An exponential distribution is obtained when $b \to \infty$ and $\alpha/b \to \beta > 0$. To see this, cancel the common factor $b^{-\alpha}$ from the numerator and denominator:

$$p_r(\alpha, b) = \frac{\exp[-\alpha \log(1 + r/b)]}{\sum_{s=1}^{R} \exp[-\alpha \log(1 + s/b)]}. \tag{4}$$

Since R is fixed, the expansion

$$\alpha \log(1 + r/b) = \frac{\alpha}{b} r + O\left(\frac{\alpha R^2}{b^2}\right) \tag{5}$$

holds uniformly over the ranks. Under the specified scaling, the remainder tends to zero and hence

$$p_r(\alpha, b) \longrightarrow \frac{e^{-\beta r}}{\sum_{s=1}^{R} e^{-\beta s}}, \tag{6}$$

recovering Eq. (1). The scaling of both parameters matters: if b increases while α is held fixed, the limiting distribution is uniform, $p_r \to 1/R$.

This limiting relationship helps interpret comparisons of the Zipf–Mandelbrot and exponential models. Whether fitted parameters for a particular corpus approach this regime requires examining their values relative to the observed rank range. The limiting argument alone makes no claim about the empirical preference between the models.

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